UNDERSTANDING OUR ENVIRONMENT AND EARTH SYSTEMS

Environmental Studies | Module I

We live within an interconnected environmental system — and as engineers, we also modify it.

S.Y. B. Tech Civil Engineering

Why Do Engineers Need to Understand the Environment?

Engineering is not separate from the environment.

Resources

Projects use land, water, energy and materials.

Changes

Projects modify natural and built environments.

Consequences

Projects can create benefits, wastes and environmental impacts.

Examples

Buildings • Dams • Roads • Water-supply systems • Wastewater treatment • Drainage

Engineering mindset

Understand the system before modifying it; anticipate direct and indirect effects.

An engineer designs WITHIN an environmental system — not outside it.

What Is Environment?

CONCEPT

The environment includes natural, built and social surroundings.

Natural environment

Air • Water • Soil • Plants • Animals • Natural ecosystems

Built environment

Buildings • Roads • Dams • Cities • Industries • Infrastructure

Social environment

People • Communities • Culture • Economy • Institutions

Environment = physical + chemical + biological

+ social conditions that surround and influence living organisms.

Components of the Environment

CONCEPT

Living and non-living components continuously interact.

Abiotic components

Air
Water
Soil and rocks
Temperature
Sunlight
Nutrients

INTERACTION

Biotic components

Plants
Animals
Microorganisms
Humans

Abiotic conditions support life
↕
Living organisms modify their surroundings

ECOSYSTEM

Interactions + energy flow + matter cycling

The Four Major Earth Systems

Each system has distinct characteristics, but none functions in isolation.

ATMOSPHERE

Gases • weather • climate

HYDROSPHERE

Oceans • rivers • groundwater • ice

INTERCONNECTED EARTH SYSTEMS

GEOSPHERE

Rocks • minerals • soil • landforms

BIOSPHERE

Plants • animals • microbes • humans

Energy Flow and Cycling of Matter

Two fundamental processes sustain Earth systems.

Energy flows

SUN
↓
Earth & ecosystems
↓
Biological processes
↓
Heat leaves the system

Energy flows in one direction.

Matter cycles

Water cycle
Carbon cycle
Nitrogen cycle
Phosphorus cycle

Matter is transformed and reused through interconnected pathways.

Engineering activities can alter the rate, pathway or location of these flows and cycles.

Human Beings and the Environment

HUMANS

Humans depend upon the environment and also drive environmental change.

Environment → Human well-being

Water
Food
Energy
Materials
Land
Ecosystem services

Human activities → Environmental change

Agriculture
Construction
Urbanization
Industry
Mining
Transportation

DEPENDENCE ↔ MODIFICATION

What Is a Man-Engineered Ecosystem?

Natural processes + human intervention + engineered infrastructure.

MAN-ENGINEERED ECOSYSTEM

Natural processes

Water flows
Biological processes
Nutrient cycles

Human intervention

Planning
Management
Resource use
Operational control

Engineered infrastructure

Buildings
Reservoirs
Treatment plants
Drainage systems

Examples: Cities • reservoirs • irrigated agriculture • urban lakes • managed wetlands • landscaped areas • wastewater treatment systems

Natural Ecosystem vs Human-Engineered Ecosystem

COMPARISON

Human engineering changes the degree and direction of environmental control.

ATTRIBUTE	NATURAL ECOSYSTEM	HUMAN-ENGINEERED ECOSYSTEM
Development	Primarily natural processes	Created or substantially modified by humans
Biodiversity	Often relatively high	Often selected or managed
Energy	Mostly natural inputs	Natural + externally supplied energy
Nutrient cycles	Largely self-regulating	Often require management
Change	Natural drivers dominate	Human decisions strongly influence change
Examples	Forest • wetland	City • reservoir • managed lake

Human-engineered ecosystems still operate according to ecological principles.

Civil Engineering as Environmental System Modification

Every major civil engineering intervention changes one or more environmental processes.

DAM

River flow • sediment • aquatic habitats

HIGHWAY

Land use • drainage • vegetation • noise

BUILDING

Land • energy • stormwater • microclimate

WATER TREATMENT

Transforms raw water into a usable resource

WASTEWATER TREATMENT

Uses physical, chemical & biological processes

URBAN DRAINAGE

Changes the movement and timing of stormwater

Civil engineers are users, designers and managers of human–environment systems.

Thinking in Systems: A Civil Engineer's Perspective

Ask not only what a project does, but what it changes — and what changes because of it.



Also consider the wider system:

Population → water demand → wastewater → solid waste → energy demand → environmental pressure

Systems thinking = Cause → Effect → Interaction → Feedback → Long-term consequence

Key Takeaways: Environment as a Connected System

Five ideas to carry forward into the rest of Environmental Studies.

1. Environment includes natural, built and social components.
2. Biotic and abiotic components interact to form ecosystems.
3. Atmosphere, hydrosphere, geosphere and biosphere are interconnected.
4. Humans both depend upon and modify environmental systems.
5. Civil engineering creates and manages human-engineered ecosystems.

Every engineering project is, in some way, an intervention in the Earth system.

THINK ABOUT IT: If a city is a human-engineered ecosystem, what resources does it consume — and what does it return to the environment?